MILITARY SPECIFICATION
BOOSTER, M125A1, LOADING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of booster, designated as Booster, M125A1, for use in high explosive ammunition.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or requests for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

Military

MIL-B-46654	- Booster M125A1, Assembly Less Booster Pellet and Lead Cup Charge

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-1167	- Ammunition Data Card

DRAWINGS

C7548175	- Box, Packing Ammunition for Booster Assembly, Details and Packing
F8594668	- Booster, M125A1, Subassembly
F8595541	- Booster, M125A1 Assembly
A8648248	- Set Up for Determining Specific Gravity
IEL-8595541	- Inspection Equipment List for

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 1390

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3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced standards and specifications.

3.2 First article sample.- (Not applicable).

3.3 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specification and the following:

3.3.1 Foreign matter.- No part or assembly shall contain chips, dirt, grease, rust, corrosion, or other foreign matter.

3.3.2 Burr.- No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define assurance terms used.

4.2 First article sample.- (Not applicable).

4.3 Inspection provisions.

4.3.1 Submission of product.- The product shall be presented in accordance with MIL-STD-105.

4.3.1.1 Lot formation.- Each inspection lot of safety adapter shall be comprised of components (except explosive components) each produced by one manufacturer, by one process, in accordance with the same drawing and revision and specifications and revisions, as applicable, and assembled on one assembly line. Each explosive component within a lot of assembled safety adapter shall be limited to one manufacturer and one interfix number.

4.3.2 Examination.- One-hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with MIL-STD-105. For classification of defects contained herein, the AQL given shall be used. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burrs and foreign matter.

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4.3.2.1 Booster subassembly, F8594668, covering a detail of Drawing F8595541.

Defect		Method of Inspection
CRITICAL		
None defined		
MAJOR	AQL = .40%	
101.	Insecure, improper or imcomplete assembly	Visual and Manual
102.	Moisture in flash hole	Visual
103.	Foreign matter in flash hole	Visual
104.	Booster damaged or defective	Visual
MINOR		
None defined		

4.3.2.2 Closing cup assembly, Drawing B8595506, covering a detail of Drawing F8595541.

CRITICAL		
None defined		
MAJOR	AQL = .40%	
101.	Explosive above flush	Gage
102.	Explosive below flush exceeds max.	Gage
MINOR		
None defined		

4.3.2.3 Booster assembly, Drawing F8595541 (prior to assembly of booster pellet and cup).

CRITICAL		
1.	Explosive material on small external threads of body.	Visual
2.	Explosive material on internal thread of booster cup.	Visual

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4.3.2.3 (Cont'd)

Defect	Method of Inspection
MAJOR AQL = .40%	
101. Closing cup charge flush or below	Gage
102. Closing cup assembly missing	Visual
103. Improper or incomplete crimp	Visual
104. Insecure or improper assembly	Manual/Visual
105. Foreign matter	Visual
MINOR	
None defined	

4.3.2.4 Booster assembly F8595541.

CRITICAL	
1. Booster armed or partially armed (see 6.1.1 and 6.1.2)	Visual
MAJOR AQL = .40%	
101. Total length exceeds max.	Gage
102. Staking missing or inadequate (if applicable)	Visual
103. Booster cup cracked or split	Visual
104. Booster cup pellet missing (an indication that booster pellet might be missing, if booster cup bottoms on shoulder of body assembly)	Visual
105. Booster pellet moves when assembly is shaken	Manual
106. Threads damaged or not full	Visual
107. Booster cup loosens when an unscrewing torque of 35 inch pounds is applied	Gage
108. Dust cover missing or loose	Visual
109. Thread sealant missing or inadequate (if applicable)	Visual
110. Foreign matter	Visual
111. Marking missing, misleading or illegible	Visual
MINOR	
None defined	

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4.3.2.5 Box, Set-up, Packing Ammunition for Boosters, Drawing C7548174, covering a detail of Drawing C7548175.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR AQL = 1.0%	
101. Box damaged so that contents are exposed or liable to become exposed	Visual
102. Sealing strip torn, badly wrinkled or incomplete	Visual
MINOR AQL = 2.5%	
201. Contents moves when shaken	Manual
202. Marking missing, misleading, incorrect or illegible	Visual

4.3.2.6 Box, Packing Ammunition for Booster, Drawing C7548175.

CRITICAL	
None defined	
MAJOR AQL = 1.0%	
101. Board broken	Visual
102. Hardware missing, broken or loose	Visual/Manual
103. Contents exposed	Visual
104. F.S.N. and D.O.D. indentification misleading, incorrect or illegible	Visual
105. Strapping missing, broken or loose	Visual/Manual
MINOR AQL = 2.5%	
201. Hardware improperly applied or engaged	Visual/Manual
202. Contents move when shaken	Manual
203. Strapping improperly applied	Visual/Manual
204. Seal missing, unsealed or improperly positioned	Visual/Manual
205. Marking misleading, incorrect or illegible	Visual

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4.3.2.7 Process control audit.-Periodic unscheduled monitoring shall be conducted for the process operations on the following assemblies to determine compliance with the requirements of the applicable drawings.

a. Booster, M125A1, subassembly, F8594668.

(1) Removal of dust seal

b. Booster, M125A1, assembly, F8595541.

(1) Insertion of closing cup assembly.

(2) Assembly of booster pellet and cup.

(3) Application of sealing compound to cup threads.

(4) Staking operation (if applicable).

4.3.3 Inspection testing.-The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Binder - lubricant content of tetryl.-A minimum of one sample (8 ounces) to a maximum of five samples (40 ounces) shall be required from each lot to determine the content of binder-lubricant of the tetryl in each lot of booster. Failure of the sample to meet the requirements specified on the applicable drawings shall be cause for rejection of the lot.

4.3.3.2 Specific gravity of booster pellet.-A sample of ten (10) booster pellets shall be selected from each lot of booster pellets to determine the specific gravity. Failure of one or more booster pellets to comply with the requirements specified on the applicable drawing shall be cause for rejection of the lot.

4.3.3.3 Inspection equipment.-The examinations and tests shall be made using equipment listed in IEL-8595541. Equipment design shall be in accordance with the code designation on the applicable IEL.

4.4 Test methods and procedures.

4.4.1 Binder-lubricant content of tetryl.-A sample of approximately 8 ounces of tetryl shall be selected and determined as follows: An accurately weighted portion of five (5) grams of the sample shall be transferred to a tared medium porosity sintered glass filtering crucible. The sample shall be extracted with four 10-milliliter portions of acetone. Each portion shall be

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4.4.1 (Cont'd)

maintained in contact with the residue for approximately five (5) minutes before applying suction. Any lumps which may persist shall be broken up. The side of the crucible shall be washed down with each 10 milliliter portion of acetone to insure complete removal of the tetryl. The residue shall be aspirated until the odor of acetone is no longer detectable. The crucible and contents shall be placed in a drying oven at 100 degrees (°) to 105° Centigrade(C) for 1/2 hour, cooled in a desiccator and weighed. The above extraction, drying and weighing procedure shall be repeated with individual 10-milliliter portions of acetone until the loss in weight on subsequent extractions totals less than 0.005 gram. The weight of the residue in the crucible shall be calculated to the percentage of total binder-lubricant in the sample to determine compliance with the requirements specified on the applicable drawing.

4.4.2 Specific gravity of booster pellet.-The representative booster pellets shall be cleaned so as to leave no sharp corners and shall then be accurately weighed on a laboratory balance. To determine the displacement, an apparatus (A8648248) consisting of a glass beaker with ground top, a piece of flat glass of sufficient size to completely cover the top of the beaker, and a semihard rubber perforated plate, pierced at each corner with sharp pointed pins so that about 1/2 inch of pin protrudes from each side, shall be used. The beaker shall be filled with clean mercury, the rubber plate inserted, and the mercury leveled off by pressing down the flat glass until it is even with the top of the beaker. The sample booster pellets shall be placed beneath the prongs of the rubber plate and the mercury leveled as before. The excess mercury shall be caught in a clean bowl. The mercury displaced by the pellet shall be calculated as follows to determine compliance with the requirement specified on the applicable drawing:

$$\text{Specific gravity} = \frac{\sigma W}{W_1}$$

where σ = density of mercury at test temperature, in grams/cubic centimeter.
W = weight of booster pellets, in grams.
W_1 = weight of displaced mercury, in grams.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, labeling, and marking shall be as specified on Drawing C7548175.

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5.2 Data cards.-Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 Armed/Unarmed boosters.

6.1.1 Armed.-Rotor is locked by the spring loaded rotor lock pin in the position in which the detonator is fully visible through the flash hole.

6.1.2 Partially armed.-Rotor not locked by both spring-loaded detents.

6.2 Ordering data.-Procurement documents should specify the title, number and date of this specification.

6.3 Supersession data.-This specification includes the requirements of Purchase Description FA-PD-MI-2592, Rev. 0 dated 5 February 1963.

Custodian:

Army - MU

Preparing activity:

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